

SI 671/721

Data Mining: Methods and Applications

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(Ph.D.) and Shan Randhawa

Session & Code	Time	Location	Lead
Lecture SI 671-001 & 101, SI 721-001	Tues 6:30 PM - 8:30 PM	1125 LCSIB	Jinseok Kim Shan Randhawa Sualeha Farid
Discussion SI 671-002 & 102	Thurs 9:00AM - 10:00AM	2270 LCSIB	Shan Randhawa
Discussion SI 671-003 & 103	Weds 11:30AM - 12:30PM	2270 LCSIB	TBA
Discussion SI 671-004 & 104	Weds 3:30PM - 4:30PM	2270 LCSIB	TBA
Discussion SI 671-005 & 105, SI 721-005	Weds 7:30PM - 8:30PM	2280 LCSIB	TBA
Discussion SI 671-006 & 106	Thurs 6:30PM - 7:30PM	2270 LCSIB	Shan Randhawa

Instructor	Sessions in Charge	Office hours and link/location (By appointment only; Zoom only)
Jinseok Kim	Lecture SI 671-001 & 101, SI 721-001	Friday 10:00 AM - 12:00 PM (15-min slots; up to 4 slots available for consecutive appointments) Appointment booking page: Link to the page Links to an external site.
Shan Randhawa	Discussion SI 671-002 & 102 Discussion SI 671-006 & 106	Thursday 10:30 Am - 12:30 PM (15-min slots) Appointment booking page: Link to the page Links to an external site.
TBA	Discussion SI 671-003 & 103 Discussion SI 671-004 & 104 Discussion SI 671-005 & 105, SI 721-005	TBA

1. Course Description

SI 671/721 is a graduate-level course on advanced topics in data mining. The course provides an overview of recent research topics in the field of data mining, state-of-the-art methods to analyze different types of datasets, and their applications to many real-world problems. The course will highlight the practical applications of data

mining instead of the theoretical foundations of machine learning and statistical computing. So, this course can be thought of as a complement to other statistical inference or machine learning courses, but not as a substitute for them.

The course materials will focus on how the information in different real-world problems can be represented as particular genres, or formats of data, and how the basic mining tasks of each genre of data can be accomplished using the state-of-the-art techniques. To this end, the course is not only suitable for masters/doctoral students who are doing research in data mining related fields, but also for students who are consumers of data mining techniques in their own disciplines, such as natural language processing, network science, human computer interaction, biomedical and health informatics, economics, social computing, sociology, and business intelligence.

2. Prerequisites

- Familiarity with Python (and libraries such as Numpy, Pandas, Scikit Learn) will be helpful. If you have a lot of programming experience but in a different language (e.g. R/C++/Matlab), you will probably be fine.
- Basic Probability and Statistics.
- It is recommended that students have some exposure to some basic ideas in statistical learning and optimization via prior courses in related areas such as Machine Learning, Data Mining, Statistical Inference, Econometrics, NLP, Computer Vision, Information Retrieval, Data Science, Social Network Analysis etc.

3. Learning Objectives

Learning objectives of this course include:

- Describe the basic principles of knowledge discovery from data.
- Perform the basic computational tasks of data mining, including pattern and association extraction, data modeling, classification, clustering, ranking, prediction, outlier detection, etc.
- Demonstrate how information in real world applications can be formulated and represented as different genres of data, such as

matrices, item sets, sequences, time series, data streams, graphs/networks, etc.

- Identify how to select appropriate data mining techniques for real-world scenarios.
- Identify the major data mining problems specific to different genres of data.
- Apply the state-of-the-art data mining techniques that solve these problems.
- Discuss the various applications of these techniques in multiple disciplines.
- Develop software development skills to deal with large-scale datasets (e.g., at least millions of data records).

4. Readings/Textbooks

There are no required textbooks for this course. Our readings will be derived from research papers published in top conferences/journals in data mining and allied areas such as KDD, WWW, ICDM, CIKM, WSDM, ICWSM, TKDE, TKDD, SIGIR.

Here's a list of optional textbooks that can be used for supplemental reading.

- Leskovec, Rajaraman, Ullman. Mining of Massive Datasets (Much of the content is available online for free: [Resource](#))
- [Links to an external site.](#)
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- Hastie, Tibshirani, Friedman. The Elements of Statistical Learning: Data Mining, Inference, and Prediction. (Available for free online: <https://web.stanford.edu/~hastie/Papers/ESLII.pdf>)

5. Course Format and Grading

The session on Tuesday will be instructor-led discussions of different data mining and data science topics. There will also be GSI-led discussion sessions (dates vary) which will cover code examples. The code files will be announced one week prior.

5.1 Grading

Grading will be based on:

Mid-term exam (in-class)	20%
Final-exam (in-class)	10%
Quiz (in-class)	10%
Three Programming Assignments	30%, split as [10%, 10%, 10%]
Final Project	30%

Conversion between percentage grades and letter grades will use the following mapping:

A+	A	A-	B+	B	B-	C+	C	C-	D+	D	D-	E	F
100	~93	~90	~87	~83	~80	~77	~73	~70	~67	~63	~60	~50	~40

5.1.1 Mid-term Exam (20%):

There will be an in-class mid-term exam. The exam will be closed book + closed notes and will test your knowledge of the concepts covered till the previous lecture (10/2). The exam questions will mostly be multiple-choice.

5.1.2 Final Exam (10%):

There will be an in-class final exam in the last class. The exam will be closed book + closed notes and will test your knowledge of the full course materials. The exam questions will mostly be multiple-choice.

5.1.3 Quiz (10%):

There will be one 30-minute short quiz after class on 11/13. It will contain multiple-choice questions based on lectures from 10/23, 10/30, and 11/6.

5.1.4 Programming Assignments (30%):

Students will be required to complete 3 data mining assignments. Each assignment requires students to program different data mining algorithms to solve real-world

problems/tasks e.g. Link prediction in social networks; sentiment classification; community detection etc.

5.1.5 Course Project (30%):

A course project is required. Individual projects are preferred but students are allowed to work in teams of 2 or 3. Grading will be adjusted accordingly, so teams of 2 or 3 will have to do proportionately more work. Course project is intended to be an open-ended project based on the interest of student(s). Typically, it will involve developing and implementing some data mining algorithms for a relevant problem. The project quality should be worthy of being published at a data mining conference. More details will be provided regarding this in the class.

The final deliverables include the software, a detailed write-up, and a presentation. The grading for the course project will be split as follows (of the 30% total):

Submission Item	Percentage (Total 30%)	Grading Rubric
Project Proposal	10%	A concrete two-page proposal, describing the project topic, objectives, expected deliverable (software package, demo, and/or a technical report), and a list of team members and their expected contribution to the project.
Final project video	10%	Students will record a 5-minute video summarizing their project. You may use slides to explain your project.
Final project deliverable and report	10%	Students are expected to submit their project deliverable, along with a detailed report. The hard deadline for this is 12/8. The report should include key observations and conclusions based on the project and suggest potential follow-up studies. Teams working on the project together must also describe individual contributions of the team members.

6. Weekly Class Schedule

Weeks	Lecture Dates	Topics	Key Dates
1	08/26	Lecture: Introduction to Data Mining (I). Lab: Python Data Science, and Linear Algebra Refresher.	
2	09/02	Lecture: Introduction to Data Mining (II). Lab: Machine Learning Modeling Refresher.	
3	09/09	Lecture: Mining Itemsets. Lab: Mining Itemsets.	HW 1 is out. Due 09/26
4	09/16	Lecture: Mining Matrix Data. Lab: Mining Matrix Data.	
5	09/23	Lecture: Mining Sequence and Text Data. Lab: Mining Sequence and Text Data.	
6	09/30	Lecture: Mining Time-series Data (I). Lab: Mining Time-series Data (I).	HW 2 is out. Due 10/24
7	10/07	Mid-term exam (In-class on Tuesday; Closed book + Closed notes) No Lab	
8	10/14	No Class on 10/14 (Fall Study Break)	

		No Lab	
9	10/21	Lecture: Mining Network Data. Lab: Mining Network Data.	HW 3 is out. Due 11/14
10	10/28	Lecture: Mining Streaming Data. Lab: Mining Streaming Data.	Final Project Proposal Due 10/31
11	11/04	Lecture: Mining Time-series Data (II). Lab: Mining Time-series Data (II).	
12	11/11	Lecture: Mining Embedded Representations + In-class Quiz (30 minutes) Lab: Embedded Representations.	
13	11/18	Mining from Experiments: Bandits and Causal Inference. Lab: (possibly) final exam preview (TBA)	
14	11/25	No Class (Thanksgiving Break) No Lab	
15	12/02	Mining Interpretable Models & Data Science Ethics (45 minutes) + Final Exam (60 minutes) on Tuesday. No Lab	

16	12/09	-	Final Project Report + Video Due on 12/09
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7. Attendance

Attendance at lectures is necessary to get the most out of the course. However, if you're ill, you may skip the class.

8. Late submission policy

Students have 72 hours of buffer grace period for the entire semester. If necessary, students may use it to submit any of the 3 homework assignments late. A student may use it all on one assignment or use a bit of it for any number of assignments. Once the buffer grace period is used up, late submissions will not be graded. Please note that the grace period can ONLY be used on the homework and NOT on the final project.

9. Academic Conduct & Course Policies

9.1. Collaboration:

UMSI strongly encourages collaboration while working on some assignments, such as homework problems and interpreting reading assignments as a general practice. Active learning is effective. Collaboration with other students in the course will be especially valuable in summarizing the reading materials and picking out the key concepts. You must, however, write your homework submission on your own, in your own words, before turning it in. If you worked with someone on the homework before writing it, you must list any and all collaborators on your written submission. Each course and each instructor may place restrictions on collaboration for any or all assignments. Read the instructions carefully and request clarification about collaboration when in doubt.

Collaboration is almost always forbidden for take-home and in class exams.

9.2. Use of AI Tools:

You are welcome to use AI Tools, e.g., Large Language Models (LLMs) such as ChatGPT on homework and the final project. However, **you must cite these tools when you turn in your work**. You must also take full responsibility for your submissions—any coding or conceptual errors will be graded as your own, regardless of whether they came from ChatGPT. In other words: proofread everything that comes out of generative AI before you turn it in. You may find it simpler to complete the homework using your own knowledge and expertise rather than AI tools. It will also be challenging to complete the in-class exams and quizzes if you are reliant on AI.

9.3. Plagiarism:

All written submissions must be your own, original work. Original work for narrative questions is not mere paraphrasing of someone else's completed answer: you must not share written answers with each other at all. At most, you should be working from notes you took while participating in a study session. Largely duplicate copies of the same assignment will receive an equal division of the total point score from the one piece of work. You may incorporate selected excerpts, statements or phrases from publications by other authors, but they must be clearly marked as quotations and must be attributed. If you build on the ideas of prior authors, you must cite their work. You may obtain copy editing assistance, and you may discuss your ideas with others, but all substantive writing and ideas must be your own, or be explicitly attributed to another. See the (Doctoral, MSI, BSI) student handbooks available on the UMSI intranet for the definition of plagiarism, resources to help you avoid it, and the consequences for intentional or unintentional plagiarism.

9.4. Reasonable accommodations:

The University of Michigan recognizes disability as an integral part of diversity and is committed to creating an inclusive and equitable educational environment for students with disabilities. Students who are experiencing a disability-related barrier should contact Services for Students with Disabilities <https://ssd.umich.edu/>; 734-763-3000 or ssdoffice@umich.edu). For students who are connected with SSD, accommodation requests can be made in Accommodate. If you have any questions or concerns please contact your SSD Coordinator or visit SSD's Current Student webpage. SSD considers aspects of the course design, course learning objects and the individual academic and course barriers experienced by the student. Further conversation with SSD, instructors, and the student may be warranted to ensure an accessible course experience. The instructional team will treat any information that you provide in as confidential a manner as possible.

9.5. Student Mental Health and Wellbeing:

The University of Michigan is committed to advancing the mental health and wellbeing of its students, while acknowledging that a variety of issues, such as strained relationships, increased anxiety, alcohol/drug problems, and depression, directly impacts students' academic performance.

If you or someone you know is feeling overwhelmed, depressed, and/or in need of support, services are available. For help, contact Counseling and Psychological Services (CAPS) at (734) 764-8312 and <https://caps.umich.edu/> during and after hours, on weekends and holidays or through its counselors physically located in schools on both North and Central Campus.

You may also consult University Health Service (UHS) at (732) 764-8320 and <https://www.uhs.umich.edu/mentalhealthsvcs>, or for alcohol or drug concerns, see www.uhs.umich.edu/aodresources.

For a more comprehensive listing of the broad range of mental health services available on campus, please visit: <https://umich.edu/~mhealth/>.

University students may experience stressors that can impact both their academic experience and their personal well-being. These may include academic pressures and challenges associated with relationships, mental health, alcohol or other drugs, identities, finances, etc. If you are experiencing concerns, seeking help is a courageous thing to do for yourself and those who care about you.

If the source of your stressors is academic, please contact me so that we can find solutions together. Ashley Ewearitt, a Counseling and Psychological Services (CAPS) counselor, is embedded in UMSI, information about how to schedule an appointment with her can be found here. For personal concerns, U-M offers a variety of resources, many which are listed on the Resources for Student Well-being webpage. You can also search for additional well-being resources on that website.